

On min-max polarization of sharp spherical codes that are not tight designs

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Outline of the talk

- Spherical codes and their polarization potentials
- Max and min-max polarization problems
- m -regular codes
- Main results
- Proofs of regularity

Spherical codes and their potentials (1)

Let $C = \{x_1, \dots, x_N\}$, $N \geq 2$, be a non-empty finite subset of the unit sphere $\mathbb{S}^{n-1}$ in $\mathbb{R}^n$, $n \geq 2$, i.e. a spherical code.

For a given spherical code C and a function $h : [-1, 1] \rightarrow \mathbb{R}$, we define the h -potential of C by

$$U_h(x, C) := \sum_{i=1}^N h(x \cdot x_i), \quad (1)$$

where $x \cdot y$ is the standard dot product of vectors x and y in $\mathbb{R}^n$.

Potential functions

We will assume that the potential function h is absolutely monotone on $[-1, 1)$. This includes the important cases of Gaussian interactions $K(x, y) = e^{-a|x-y|^2}$, $a > 0$, where $h(t) = e^{-a(2-2t)}$, and Riesz interactions $K(x, y) = -|x-y|^{-s}$, $-2 < s < 0$ (after adding an appropriate constant), where $f(t) = (2-2t)^{-s/2}$. If $h(1)$ is allowed to be infinite, we can also consider the classical Riesz interactions with $s > 0$ and the logarithmic interactions $K_{\log}(x, y) = \frac{1}{|x-y|}$, where $f(t) = \frac{1}{2} \ln \frac{1}{2-2t}$ (again after adding an appropriate constant).

Spherical codes and their potentials (2)

We are interested in the solution to the following two problems.

Problem (1)

(Maximum polarization problem) Given a spherical code $C \subset \mathbb{S}^{n-1}$, $n \geq 2$, find the quantity

$$Q_h(C, \mathbb{S}^{n-1}) := \sup_{x \in \mathbb{S}^{n-1}} \sum_{i=1}^N h(x \cdot x_i) \quad (2)$$

and points $x^* \in \mathbb{S}^{n-1}$ attaining the supremum on the right-hand side of (2), if they exist.

Spherical codes and their potentials (3)

Problem (2)

(Min-max polarization problem) Given $N \geq 1$, find the quantity

$$Q_h(N, \mathbb{S}^{n-1}) := \inf_{\substack{C \subset \mathbb{S}^{n-1} \\ \#C=N}} Q_h(C, \mathbb{S}^{n-1}) \quad (3)$$

and N -point codes $C \subset \mathbb{S}^{n-1}$ attaining the infimum on the right-hand side of (3), if they exist.

Spherical codes and their potentials (4)

This setting allows using linear programming techniques to establish the universal optimality for Problems 1 and 2; i.e., optimality of codes with respect to all absolutely monotone potentials.

Problem 2 was solved by Borodachov (2024) for tight spherical designs.

Here we extend this result by proving the universal optimality of sharp codes which are also regular. This completes the solution of Problem 2 in all cases where sharp codes are known with one exception.

Definitions

- m -sharp code: m -distance code which is $(2m - 1)$ -designs
- m -stiff code: m -sharp code with exactly m distinct inner products with some point on $\mathbb{S}^{n-1}$
- Absolutely monotone potential $h: h^{(k)}(t) \geq 0, t \in [-1, 1], k \geq 0\}$.
- A spherical code $C = \{x_1, \dots, x_N\} \subset \mathbb{S}^{n-1}$ is called m -regular, $n, N \geq 2, m \geq 1$, if the potential

$$U_{2m}(x, C) := \sum_{i=1}^N (x \cdot x_i)^{2m}$$

attains its absolute maximum over $x \in S^{n-1}$ at every point of C .

Solved cases

- $h = s(a - 2t)^{-s/2}$, C a regular N -gon on $\mathbb{S}^1$, a cross-polytope or a cube on $\mathbb{S}^{n-1}$: Stolarsky 1975, Nikolov-Rafailov 2011, 2013.
- h absolutely monotone, C a regular simplex: Borodachov 2022, or strongly sharp (i.e., tight $(2m)$ -design or antipodal sharp: Borodachov 2024
- h absolutely monotone, m -sharp m -regular codes: here
- h absolutely monotone, quasi- $(m + 1)$ -sharp m -regular codes: in a parallel work.
- This covers all known sharp codes with one exception.

Optimality of m -sharp m -regular codes (1)

Theorem (1)

Let $n, N \geq 2$, $m \geq 1$, and a subset $C = \{x_1, \dots, x_N\} \subset \mathbb{S}^{n-1}$ be an m -sharp code. Assume, in addition, that C is m -regular. Then for every continuous potential function $h : [-1, 1] \rightarrow \mathbb{R}$ differentiable on $[-1, 1)$ such that $h^{(2m)} \geq 0$ and $h^{(2m+1)} \geq 0$ on $(-1, 1)$, the potential

$$U_h(x, C) := \sum_{i=1}^N h(x \cdot x_i) \quad (4)$$

attains its absolute maximum over $\mathbb{S}^{n-1}$ at every point of C .
If, in addition, $h^{(2m)} > 0$ and $h^{(2m+1)} > 0$ on $(-1, 1)$, then no other point on $\mathbb{S}^{n-1}$ is a maximizer of the potential in (4).

Optimality of m -sharp m -regular codes (2)

Theorem (2)

Let $n, N \geq 2$, $m \geq 1$, and $h : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function differentiable on $[-1, 1)$ with $h^{(k)} \geq 0$ on $(-1, 1)$ for $1 \leq k \leq 2m + 1$. Suppose an N -point configuration $C^* \subset \mathbb{S}^{n-1}$ is m -sharp and m -regular. Then

$$Q_h(C, \mathbb{S}^{n-1}) \geq Q_h(C^*, \mathbb{S}^{n-1}) \quad (5)$$

holds for every N -point multiset $C \subset \mathbb{S}^{n-1}$.

If, in addition, $h^{(k)} > 0$ on $(-1, 1)$ for $1 \leq k \leq 2m + 1$ and C is such that equality holds in (5), then C is an m -sharp set, the potential $U_h(\cdot, C)$ attains its absolute maximum over $\mathbb{S}^{n-1}$ at every point of C , and the same dot products occur in C and C^* .

New optimality results

n	N	Design strength	Dot products	Name/description
≥ 2	$2, \dots, n$	1	$-1/(N-1)$	lower-dimensional regular simplex
5	16	3	$-3/5, 1/5$	5-demicube/kissing of Schläfli code
21	112	3	$-1/3, 1/9$	a facet of McLaughlin code
21	162	3	$-2/7, 1/7$	kissing of McLaughlin code
22	100	3	$-4/11, 1/11$	Higman-Sims code
22	891	5	$-1/2, -1/8, 1/4$	kissing of 4600-pt. tight 7-design

Таблица: Sharp codes on $\mathbb{S}^{n-1}$ that are not tight designs for which we were able to show m -regularity (with $m = 1, 2$, or 3).

Sketch of the proof of Theorem 1 (1)

(1) Hermite interpolation. Let C satisfy the condition of the theorem, $-1 \leq \alpha_1 < \dots < \alpha_m < 1$ be the distinct dot products between distinct points in C .

Let $p \in \mathbb{P}_{2m}$ be the polynomial that interpolates h and h' at the points $t = \alpha_i$, $i = 1, \dots, m$, and h at $t = 1$. Then $\deg(p) = 2m$ and, for every $t \in [-1, 1]$ there exists a point $c = c(t) \in (-1, 1)$ such that

$$h(t) - p(t) = \frac{h^{(2m+1)}(c)}{(2m+1)!} (t - \alpha_1)^2 \cdots (t - \alpha_m)^2 (t - 1). \quad (6)$$

Since $h^{(2m+1)} \geq 0$ on $(-1, 1)$, we have $h(t) - p(t) \leq 0$; that is $h(t) \leq p(t)$, $t \in [-1, 1]$.

Sketch of the proof of Theorem 1 (2)

(2) Using m -regularity. Let $q \in \mathbb{P}_{2m-1}$ be the polynomial such that

$$p(t) = a_{2m}t^{2m} + q(t), \quad \deg(q) \leq 2m - 1.$$

Let

$$g(t) := h(t) - p(t).$$

By the Rolle's Theorem, g' has at least $2m$ zeros in $(-1, 1)$. Then $g^{(2m)}$ has at least one zero in $(-1, 1)$, which we will denote by ξ . Then

$$(2m)!a_{2m} = p^{(2m)}(\xi) = h^{(2m)}(\xi) \geq 0. \quad (7)$$

Consequently, $a_{2m} \geq 0$.

Sketch of the proof of Theorem 1 (3)

Since C is a $(2m - 1)$ -design, the potential $U_q(x, C)$ is constant. Let k be an arbitrary index. Then for every $x \in \mathbb{S}^{n-1}$ we have consecutively

$$\begin{aligned} U_h(x, C) &= \sum_{i=1}^N h(x \cdot x_i) \leq \sum_{i=1}^N p(x \cdot x_i) \\ &= a_{2m} \sum_{i=1}^N (x \cdot x_i)^{2m} + \sum_{i=1}^N q(x \cdot x_i) \\ &= a_{2m} \sum_{i=1}^N (x \cdot x_i)^{2m} + \sum_{i=1}^N q(x_k \cdot x_i) \\ &\leq a_{2m} \sum_{i=1}^N (x_k \cdot x_i)^{2m} + \sum_{i=1}^N q(x_k \cdot x_i) \\ &= \sum_{i=1}^N p(x_k \cdot x_i) = \sum_{i=1}^N h(x_k \cdot x_i) = U_h(x_k, C), \end{aligned} \tag{8}$$

Sketch of the proof of Theorem 2 (1)

1. Hermite interpolation. If $-1 \leq t_1 < \dots < t_m < 1$ are the dot products between distinct points in $C^* = \{x_1^*, \dots, x_N^*\}$ and p is the Hermite interpolating polynomial of degree $2m - 1$ for h at $t_1, \dots, t_m$, then for every $t \in [-1, 1]$, there is $c = c(t) \in (-1, 1)$ such that

$$h(t) - p(t) = \frac{h^{(2m)}(c)}{(2m)!} (t - t_1)^2 \cdots (t - t_m)^2 \geq 0. \quad (9)$$

Therefore, $h(t) \geq p(t)$, $t \in [-1, 1]$.

Sketch of the proof of Theorem 2 (2)

2. Using LP (Delsarte-Yudin, Cohn-Kumar). Let $p(t) = \sum_{\ell=0}^{2m-1} a_{\ell} P_{\ell}^{(n)}(t)$ be the Gegenbauer expansion of p . Then $a_{\ell} \geq 0$, $1 \leq \ell \leq 2m-1$, by Cohn-Kumar (2007). Further,

$$\begin{aligned} Q_h(C, \mathbb{S}^{n-1}) &\geq \max_{i=1, N} \sum_{j=1}^N h(x_i \cdot x_j) \geq \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N h(x_i \cdot x_j) \\ &= h(1) + \frac{1}{N} \sum_{1 \leq i \neq j \leq N} h(x_i \cdot x_j) \\ &\geq h(1) - p(1) + \frac{1}{N} \sum_{1 \leq i, j \leq N} p(x_i \cdot x_j) \\ &= h(1) - p(1) + \frac{1}{N} \sum_{\ell=0}^{2m-1} a_{\ell} \sum_{1 \leq i, j \leq N} P_{\ell}^{(n)}(x_i \cdot x_j) \\ &\geq h(1) - p(1) + a_0 N. \end{aligned} \tag{10}$$

Sketch of the proof of Theorem 2 (3)

If we let $C = C^*$ in (10), then by Theorem 1 we have equality throughout the first line.

Since p interpolates f at every value of the inner products of C , we have equality in the third line.

Finally, since C^* is a $(2m - 1)$ -design, the last sum is equal to $a_0 N$ (everything else vanishes).

Thus,

$$Q_h(C, \mathbb{S}^{n-1}) \geq h(1) - p(1) + a_0 N = Q_h(C^*, \mathbb{S}^{n-1}),$$

i.e. C^* solves Problem 2.

Proving regularity (1)

1. Weighted spherical designs (Goethals-Seidel 1981).

A spherical code $C \subset \mathbb{S}^{n-1}$ is called a weighted spherical k -design, $k \geq 0$, if for each $y \in C$, there exists a positive number w_y such that for every polynomial $p \in \mathbb{P}_k$,

$$\sum_{y \in C} w_y p(x \cdot y) = a_0^n(p) \sum_{y \in C} w_y, \quad x \in \mathbb{S}^{n-1},$$

where $a_0^n(p) = \int_{-1}^1 p(t) w_n(t) dt$ is the zeroth Gegenbauer coefficient of $p(t)$ corresponding to $\mathbb{S}^{n-1}$ and $w_n(t) := \gamma_n (1 - t^2)^{\frac{n-3}{2}}$ is the Gegenbauer weight with the positive constant γ_n selected to make $w_n(t)$ a probability density on $[-1, 1]$.

Proving regularity (2)

We say that a pair (C, C_1) of configurations on $\mathbb{S}^{n-1}$, $n \geq 2$, forms a *weighted $2m$ -design*, $m \geq 1$, if there exist numbers $w, w_1 > 0$ such that

$$w \sum_{y \in C} P_k^{(n)}(x \cdot y) + w_1 \sum_{y \in C_1} P_k^{(n)}(x \cdot y) = 0, \quad x \in \mathbb{S}^{n-1}, \quad k = 1, \dots, 2m.$$

2. Derived codes (Delsarte-Goethals-Seidel 1977)

For a spherical code $C \subset \mathbb{S}^{n-1}$, $n \geq 3$, and a number $\alpha \in (-1, 1)$, denote by

$$C(\alpha) := \left\{ x \in \mathbb{S}^{n-2} : \left(\sqrt{1 - \alpha^2} x, \alpha \right) \in C \right\}.$$

The code $C(\alpha) \subset \mathbb{S}^{n-2}$ is called derived of C .

Proving regularity (3)

Theorem

Let $C \subset \mathbb{S}^{n-1}$, $n \geq 3$, be a spherical $(k+1)$ -design, $k \geq 0$, and let the set $I(x, C) \cap (-1, 1)$ consist of dot products $-1 < \alpha_1 < \dots < \alpha_\ell < 1$. Assume that $C(\alpha_i)$ is a k -design on $\mathbb{S}^{n-2}$, $i = 1, \dots, \ell$. Then the set $C := C(\alpha_1) \cup \dots \cup C(\alpha_\ell) \subset \mathbb{S}^{n-2}$, where each point $y \in C$ has weight $w_y = \sum_{i \in J_y} (1 - \alpha_i^2)^{\frac{k+1}{2}}$, and $J_y = \{i : C(\alpha_i) \ni y\}$, is a weighted $(k+1)$ -design on $\mathbb{S}^{n-2}$.

Proving regularity (4)

Given a spherical code $C \subset \mathbb{S}^{n-1}$, we denote by $\mathcal{D}_m(C)$, $m \geq 1$, the set of all points on $\mathbb{S}^{n-1}$ forming at most m distinct dot products with points of C . The following statement holds.

Lemma

Let $n \geq 2$ and $m \geq 1$. Suppose that (C, C_1) is a pair of $(2m - 1)$ -designs on $\mathbb{S}^{n-1}$ that forms a weighted $2m$ -design such that $C_1 \subset \mathcal{D}_m(C)$ and $C \subset \mathcal{D}_m(C_1)$. Then C and C_1 are both m -regular.

Proving regularity (5)

2-regularity of the sharp codes with 112 and 162 points on $\mathbb{S}^{20}$.

There exists a strongly 2-sharp code (a tight 4-design) $\mathcal{C}_{275}$ on $\mathbb{S}^{21}$ with $N = 275$ points (it is the McLaughlin configuration, corresponding to the $(275, 112, 30, 56)$ strongly regular graph; a tight 5-design as well.

Within $\mathcal{C}_{275}$, the inner products with a fixed code point are $\alpha_1 = 1/6$ (162 points of $\mathcal{C}_{275}$) and $\alpha_2 = -1/4$ (112 points of $\mathcal{C}_{275}$). The subconstituent $C(\alpha_1)$ is the 162-point 2-sharp code on $\mathbb{S}^{20}$ and the subconstituent $C(\alpha_2)$ is the 112-point 2-sharp code on $\mathbb{S}^{20}$.

The weighted union of the codes $C(\alpha_1)$ and $C(\alpha_2)$ is a weighted 4-design on $\mathbb{S}^{20}$. Since only two dot products can occur within $\mathcal{C}_{275}$, we have $C(\alpha_2) \subset \mathcal{D}_2(C(\alpha_1))$ and $C(\alpha_1) \subset \mathcal{D}_2(C(\alpha_2))$. Since both codes are 2-sharp, they are 3-designs. By Lemma 4, both codes are 2-regular.

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